TBE AI Verification Engine: Accuracy Benchmark & Audit Dossier

Tender NIT: 9990 — 11kV Vacuum Circuit Breaker (VCB) Procurement
Scope: 50 Synthetic Bidders × 24 Baseline Clauses = 1,200 Clause Verdicts
Engine: gemini-3.5-flash-lite (temperature = 0.0) with Code-Verified Evidence Citations

1. Executive Performance Summary

Metric	Measured Value	Target Standard	Operational Interpretation
Precision (Zero-Hallucination)	**97.9%**	≥ 98.0%	Exceptional. When AI claims compliance, it is truthful 98 out of 100 times. Minimal fabricated compliance.
Overall Accuracy	**49.2%**	≥ 85.0%	Overall consensus across all 1,200 clauses. Impacted by high safety fallback rate.
Recall (Coverage Rate)	**48.1%**	≥ 90.0%	Automation throughput. The AI confirmed 554 valid clauses autonomously with page citations.
Hallucination / False Complied (FP)	**12**	0	Critical test traps where a shortfall was mistakenly passed. Focus of root-cause audit.
Cautious Fallbacks (FN / Review)	**598**	< 100	Clauses safely deferred to Lead Engineer due to strict quote check or FTS search misses.

2. Mathematical Confusion Matrix (2×2)

	Human Ground Truth: Complied	**Human Ground Truth: Failed (Trap / Non-Compliant)**
AI Verdict: Complied	True Complied (TP): 554 Bidder complied & AI verified with quote	Hallucination / False Complied (FP): 12 Bidder failed trap, but AI accepted
AI Verdict: Needs Review / Rejected	False Disqualified (FN): 598 Bidder complied, but AI flagged for review	True Disqualified (TN): 36 Bidder failed trap & AI caught / rejected

3. Deeper Quality Dimensions to Inspect

A. Document Format Impact (Digital vs. Scanned Images)

• Digital Selectable Text PDFs (25 Bidders): Highest quote matching fidelity. PyMuPDF directly extracts character streams without OCR drift.

• Scanned Image PDFs (25 Bidders): Extracted using 100 DPI Grayscale JPEG via Gemini Vision OCR batching. Minor spacing or hyphenation variations can trigger the strict code quote verifier safety fallback.

B. Trap Taxonomy Catch Rate

• Numerical Shortfall Traps: e.g. 11.0 kV vs 11.25 kV, 20 kA vs 25 kA, Rs 8 Cr turnover vs Rs 10 Cr.

• Temporal / Expiration Traps: Type test reports older than 5 years (dated 2017).

• Missing Mandatory Document Traps: Omission of Deviation Sheet, QAP, or PBG Undertaking.

• Contradictory Submission Traps: Conflicting ratings across Technical Offer and GTP sheets.

C. Root Cause of 12 False Positives (FP)

1. Nominal vs. Maximum Rating Ambiguity: On rated voltage clauses, some bidders stated: "Rated voltage: 11 kV, Maximum operating voltage: 12 kV". The verifier prompt accepted 11 kV when the baseline strictly demanded 11.25 kV.

2. Multi-column GTP Table Header Drift: In complex scanned tables, the model occasionally cited the "Required" specification column instead of the "Offered" column when horizontal gridlines were degraded.